

Question 1

Examples of custom commands: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, $\text{trace}(\mathbf{H}) = p$. Code example:

Code Listing 1: Smoothed estimation of mean trajectory

```
1 # Compute the optimally-smoothed mean
2 lambdaOpt <- 10*logLambda[which(estimationError == min(estimationError))]
3 smoothOperatorOpt <- fdPar(bSplineBasis, int2Lfd(2), lambdaOpt)
4 smoothSpectrumOpt <- smooth.basis(logFreq, rawData, smoothOperatorOpt)
5 smoothMeanOpt <- PhiMatrix %*% mean.fd(smoothSpectrumOpt$fd)$coefs
6
7 # Plot data along unsmoothed mean and optimally-smoothed mean
8 matplot(logFreq, rawData, type="l", col=brewer.pal(8, "Set2"), xlab="Log
  ↪ frequency", ylab="Log power spectrum", main="Raw data with
  ↪ optimally-smoothed and\n unsmoothed means")
9 lines(logFreq, unsmoothedMean, lwd=2.5, col="firebrick3")
10 lines(logFreq, smoothMeanOpt, lwd=2.5, col="blue3")
```

Theorem (Leibniz integral rule). Let $f(x, t)$ and its partial derivative $\frac{\partial}{\partial x}f(x, t)$ be continuous in x and t in some region of the (x, t) plane that includes $a(x) \leq t \leq b(x)$ and $x_0 \leq x \leq x_1$. Suppose also that the functions $a(x)$ and $b(x)$ are continuous and have continuous derivatives for $x_0 \leq x \leq x_1$. Then, for $x_0 \leq x \leq x_1$, we have:

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, t) dt = f(x, b(x)) \cdot \frac{d}{dx} b(x) - f(x, a(x)) \cdot \frac{d}{dx} a(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x, t) dt$$

If $a(x) = a$ and $b(x) = b$, where a and b are constants, then the above reduces to:

$$\frac{d}{dx} \int_a^b f(x, t) dt = \int_a^b \frac{\partial}{\partial x} f(x, t) dt$$

References

- [1] Last name, first name. *Book Title*. Publisher, Year. Print.
- [2] Last name, first name. "Webpage Title". Website name, Organization name. Online; accessed Month Date, Year. www.URLhere.com